E-COMMERCE TECHNOLOGIES

PRACTICAL RECORD

Selected Experiments: 1, 2, 3, 4 and 6

Name: _____
Register Number: _____
Class / Semester: MCA _____
College: _____

Simple HTML • CSS • JavaScript

EXPERIMENT 1

Create a Timeline of EDI and E-Commerce Evolution

Date: _____

AIM

To create an interactive timeline showing the important stages in the evolution of Electronic Data Interchange (EDI) and E-Commerce.

PROCEDURE

1. Open a text editor and create a new file named crm.html.
2. Add the basic HTML structure and a heading for the E-CRM customer touchpoints.
3. Display the five customer journey stages: Awareness, Consideration, Purchase, Support and Retention.
4. Create an input field to enter the customer name.
5. Add a selection field to choose the customer interaction stage.
6. Create a notes field to describe the customer interaction.
7. Add Add Interaction and Clear List buttons with an area to display the interaction log.
8. Create a style.css file and link it to the HTML page. Apply suitable styles to the form and log.
9. Create a script.js file to validate the entries, add interactions to the log and clear the list.
10. Save all files and open crm.html in a browser. Enter sample details and test both buttons.

CODING

timeline.html

```
<!DOCTYPE html>
<html>
<head>
<title>EDI Timeline</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<h1>EDI and E-Commerce Evolution</h1>
<input id="searchBox" placeholder="Enter year or keyword">
<button onclick="searchEvents()">Search</button>
<button onclick="showAll()">Show All</button>
<p id="message">Explore the milestones.</p>
<div class="event">1960s - Early EDI</div>
<div class="event">1970s - EDI Standards</div>
<div class="event">1980s - Business Networks</div>
<div class="event">1990s - Commercial Internet</div>
<div class="event">2000s - Online Marketplaces</div>
<div class="event">2010s - Mobile Commerce</div>
<div class="event">2020s - Connected Commerce</div>
<script src="script.js"></script>
</body>
</html>
```

style.css

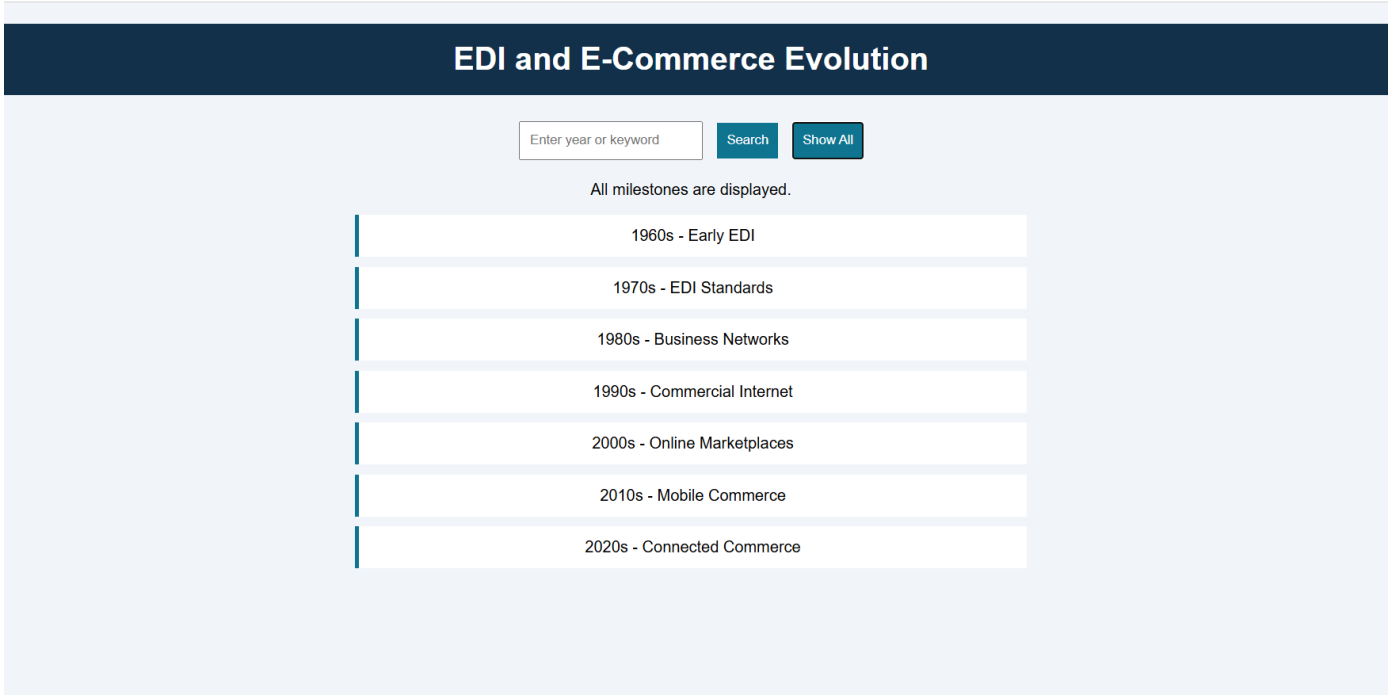
```
body {
    font-family: Arial;
    background: #f1f5f9;
    text-align: center;
}
h1 {
    background: #12304a;
```

```
    color: white;
    padding: 18px;
}
input, button {
    padding: 10px;
    margin: 5px;
}
button {
    background: #0e7490;
    color: white;
    border: 0;
}
.event {
    background: white;
    padding: 12px;
    margin: 10px auto;
    max-width: 650px;
    border-left: 4px solid #0e7490;
}
```

script.js

```
function searchEvents() {
    var word = document.getElementById("searchBox").value.toLowerCase();
    var items = document.getElementsByClassName("event"), count = 0;
    for (var i = 0; i < items.length; i++) {
        if (items[i].innerText.toLowerCase().includes(word)) {
            items[i].style.display = "block"; count++;
        } else items[i].style.display = "none";
    }
    document.getElementById("message").innerText = count + " milestone(s) found.";
}
function showAll() {
    var items = document.getElementsByClassName("event");
    for (var i = 0; i < items.length; i++) items[i].style.display = "block";
    document.getElementById("searchBox").value = "";
    document.getElementById("message").innerText = "All milestones are displayed.";
}
```

OUTPUT



RESULT

Thus, the interactive timeline of EDI and E-Commerce evolution was created and tested successfully.

EXPERIMENT 2

Simulate B2B, B2C and C2C Business Models through Roleplay

Date: _____

AIM

To demonstrate the B2B, B2C and C2C business models using simple examples of buyers and sellers.

PROCEDURE

11. Open a text editor and create a new file named timeline.html.
12. Add the basic HTML structure and give the page a suitable title.
13. Create a heading for the EDI and E-Commerce Evolution timeline.
14. Add a search box with Search and Show All buttons.
15. Create timeline cards for the seven milestones from the 1960s to the 2020s.
16. Create a separate style.css file and link it to the HTML file.
17. Style the heading, search controls and timeline cards using CSS.
18. Create a script.js file and link it before the closing body tag.
19. Write JavaScript functions to display matching milestones and show all milestones.
20. Save all three files in the same folder and open timeline.html in a web browser.

CODING

models.html

```
<!DOCTYPE html>
<html>
<head>
<title>Business Models</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<h1>E-Commerce Business Models</h1>
<button onclick="showModel('B2B')">B2B</button>
<button onclick="showModel('B2C')">B2C</button>
<button onclick="showModel('C2C')">C2C</button>
<div id="result">Select a business model.</div>
<script src="script.js"></script>
</body>
</html>
```

style.css

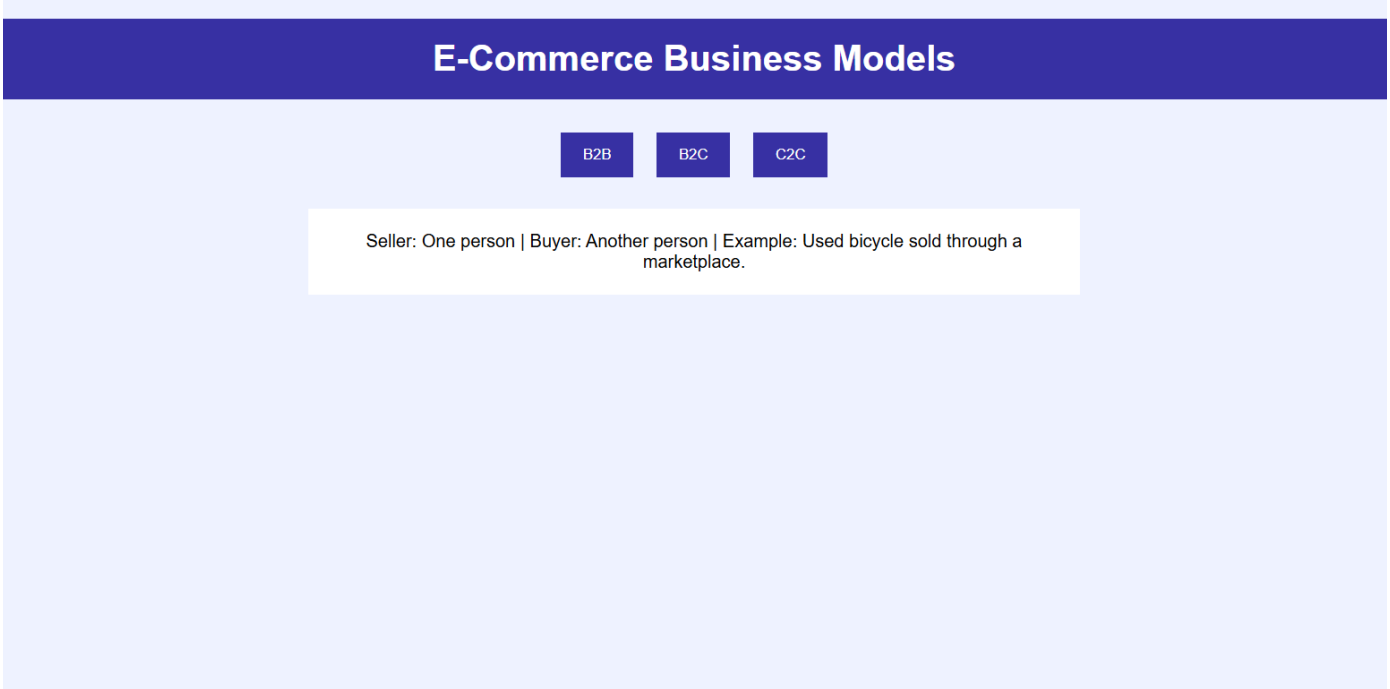
```
body {
    font-family: Arial;
    background: #eef2ff;
    text-align: center;
}
h1 {
    background: #3730a3;
    color: white;
    padding: 18px;
}
button {
    padding: 12px 20px;
    margin: 8px;
    background: #3730a3;
    color: white;
```

```
    border: 0;
}
#result {
    background: white;
    padding: 20px;
    margin: 20px auto;
    max-width: 650px;
}
```

script.js

```
function showModel(model) {
    var text = "";
    if (model == "B2B") {
        text = "Seller: Wholesaler | Buyer: Retailer | Example: 100 laptops supplied to a shop.";
    } else if (model == "B2C") {
        text = "Seller: Online store | Buyer: Customer | Example: Customer buys a mobile phone.";
    } else {
        text = "Seller: One person | Buyer: Another person | Example: Used bicycle sold through a
marketplace.";
    }
    document.getElementById("result").innerText = text;
}
```

OUTPUT



RESULT

Thus, the B2B, B2C and C2C business models were demonstrated successfully.

EXPERIMENT 3

Design a Digital Marketing Strategy for an Online Business

Date: _____

AIM

To prepare a simple digital marketing campaign by entering product, target audience, goal, budget and marketing channels.

PROCEDURE

21. Create input fields for product, target audience and monthly budget.
22. Provide a campaign goal and marketing channel checkboxes.
23. Add Create Summary and Clear buttons.
24. Use JavaScript to display the campaign details and clear the form.
25. Open a text editor and create a new file named models.html.
26. Add the basic HTML structure and set the page title.
27. Create a heading for the B2B, B2C and C2C business models.
28. Add three buttons labelled B2B, B2C and C2C.
29. Open a text editor and create a new file named marketing.html.
30. Add the basic HTML structure and a heading for the digital marketing strategy.
31. Create input fields for the product name, target audience and monthly budget.
32. Add a selection field for the campaign goal.
33. Add checkboxes for SEO, Social Media and Email marketing channels.
34. Create Create Summary and Clear buttons.
35. Create a style.css file and link it to the HTML page.
36. Style the form, labels, input fields and buttons using CSS.
37. Create a script.js file with functions to display the campaign summary and clear the form.
38. Save all files and open marketing.html in a browser. Enter the details and click Create Summary.

CODING

marketing.html

```
<!DOCTYPE html>
<html>
<head>
<title>Digital Marketing Plan</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<h1>Digital Marketing Strategy</h1>
<input id="product" placeholder="Product or service">
<input id="audience" placeholder="Target audience">
<select id="goal"><option>Increase sales</option><option>Brand awareness</option></select>
<input id="budget" type="number" placeholder="Monthly budget">
<p><label><input type="checkbox" name="channel" value="SEO"> SEO</label>
<label><input type="checkbox" name="channel" value="Social Media"> Social Media</label>
<label><input type="checkbox" name="channel" value="Email"> Email</label></p>
<button onclick="createPlan()">Create Summary</button>
<button onclick="clearPlan()">Clear</button>
<div id="summary">Campaign summary appears here.</div>
<script src="script.js"></script>
</body>
</html>
```

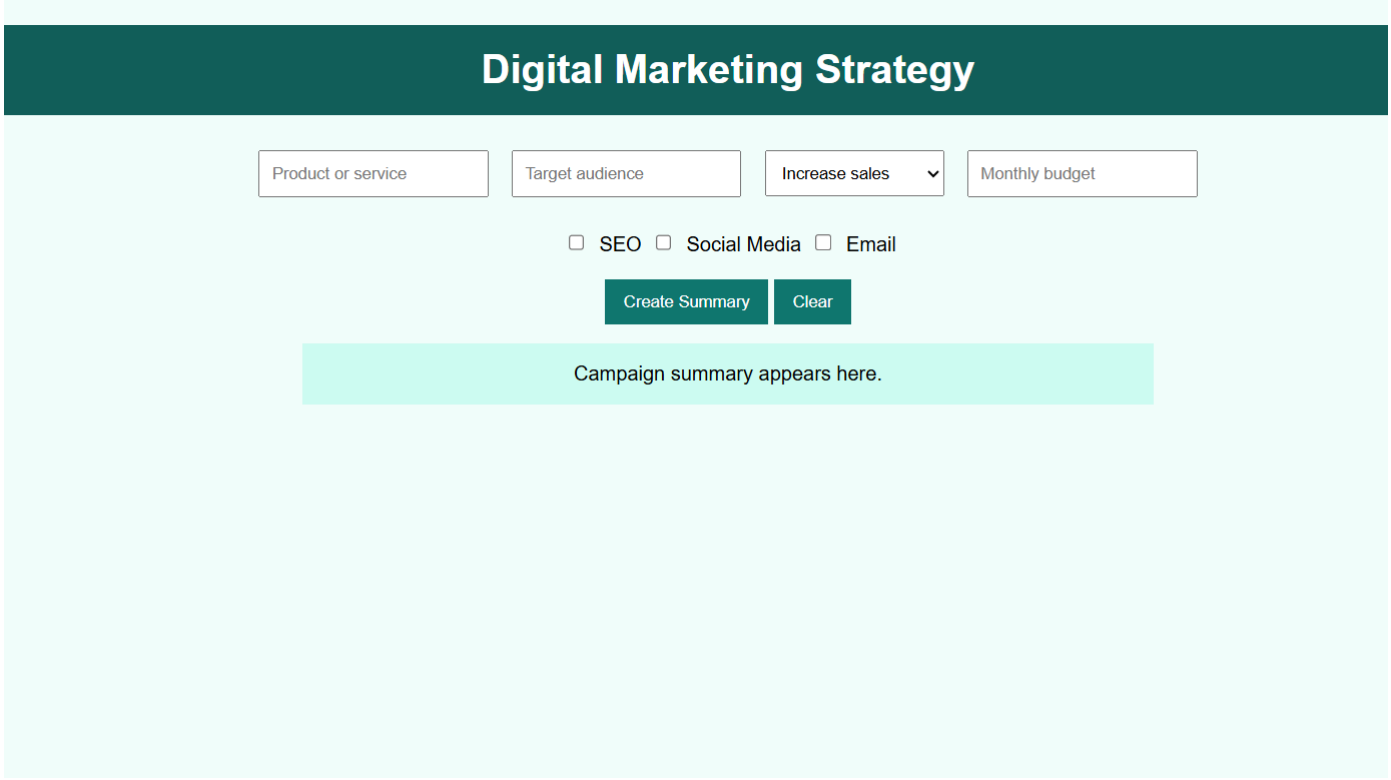
style.css

```
body {
    font-family: Arial;
    background: #f0fdfa;
    text-align: center;
}
h1 {
    background: #115e59;
    color: white;
    padding: 18px;
}
input, select {
    padding: 9px;
    margin: 7px;
}
button {
    padding: 10px 15px;
    background: #0f766e;
    color: white;
    border: 0;
}
#summary {
    background: #ccfbf1;
    padding: 15px;
    margin: 15px auto;
    max-width: 650px;
    white-space: pre-line;
}
```

script.js

```
function createPlan() {
    var channels = document.getElementsByName("channel"), chosen = [];
    for (var i = 0; i < channels.length; i++) {
        if (channels[i].checked) chosen.push(channels[i].value);
    }
    document.getElementById("summary").innerText =
        "Product: " + document.getElementById("product").value +
        "\nAudience: " + document.getElementById("audience").value +
        "\nGoal: " + document.getElementById("goal").value +
        "\nBudget: Rs. " + document.getElementById("budget").value +
        "\nChannels: " + chosen.join(", ");
}
function clearPlan() {
    document.getElementById("product").value = "";
    document.getElementById("audience").value = "";
    document.getElementById("budget").value = "";
    document.getElementById("summary").innerText = "Campaign summary appears here.";
    var boxes = document.getElementsByName("channel");
    for (var i = 0; i < boxes.length; i++) boxes[i].checked = false;
}
```

OUTPUT



RESULT

Thus, a simple digital marketing strategy planner was created and tested successfully.

EXPERIMENT 4

Compare Token-Based vs Traditional Payment Methods via Simulation

Date: _____

AIM

To compare traditional card payment and token-based payment methods and calculate an illustrative service fee.

PROCEDURE

39. Open a text editor and create a new file named marketing.html.
40. Add the basic HTML structure and a heading for the digital marketing strategy.
41. Create input fields for the product name, target audience and monthly budget.
42. Add a selection field for the campaign goal.
43. Add checkboxes for SEO, Social Media and Email marketing channels.
44. Create Create Summary and Clear buttons.
45. Create a style.css file and link it to the HTML page.
46. Style the form, labels, input fields and buttons using CSS.
47. Create a script.js file with functions to display the campaign summary and clear the form.
48. Save all files and open marketing.html in a browser. Enter the details and click Create Summary.

CODING

payments.html

```
<!DOCTYPE html>
<html>
<head>
<title>Payment Comparison</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<h1>Payment Method Comparison</h1>
<table>
<tr><th>Traditional</th><th>Token-Based</th></tr>
<tr><td>Uses card details</td><td>Uses a substitute token</td></tr>
<tr><td>Card details may be exposed</td><td>Reduces exposure of card number</td></tr>
</table>
<input id="amount" type="number" placeholder="Purchase amount">
<select id="method"><option>Traditional</option><option>Token-Based</option></select>
<button onclick="calculate()">Calculate</button>
<p id="result">Result appears here.</p>
<script src="script.js"></script>
</body>
</html>
```

style.css

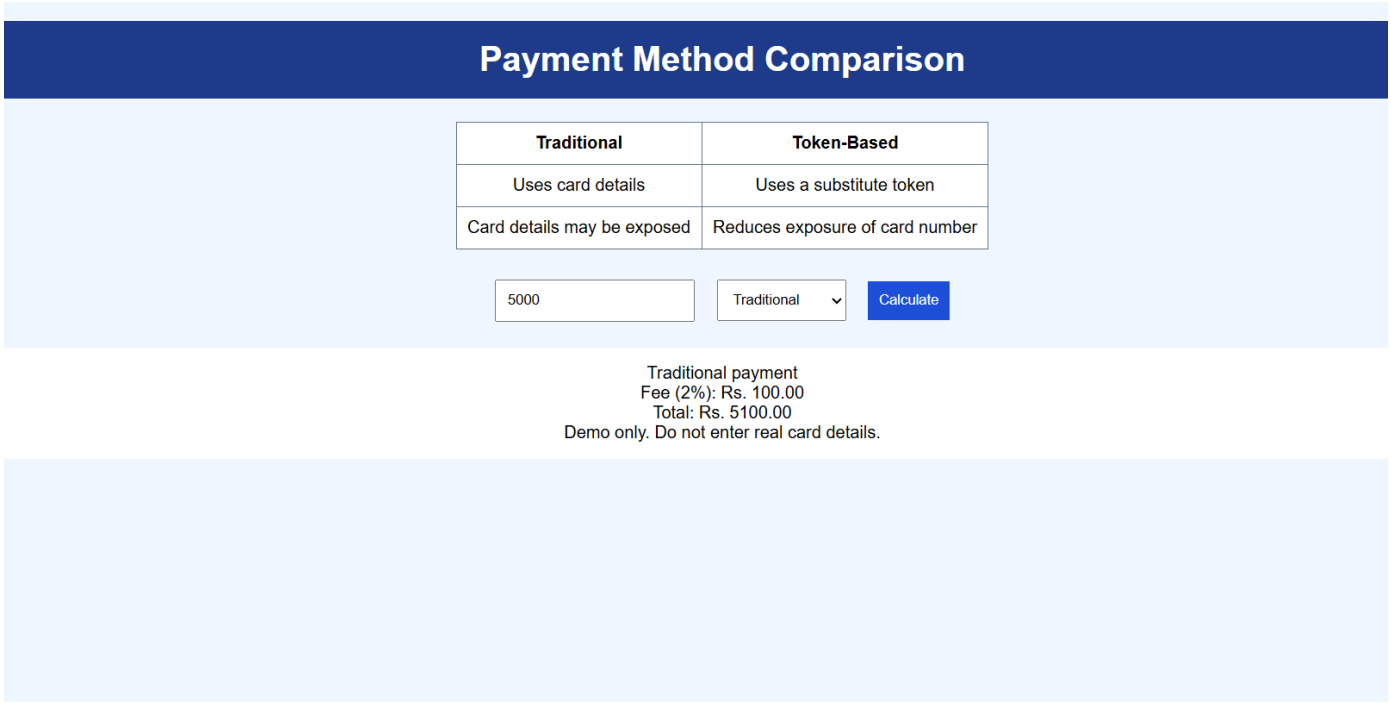
```
body {
    font-family: Arial;
    background: #eff6ff;
    text-align: center;
}
h1 {
    background: #1e3a8a;
    color: white;
    padding: 18px;
```

```
}
table {
    margin: 20px auto;
    background: white;
    border-collapse: collapse;
}
td, th {
    border: 1px solid #64748b;
    padding: 10px;
}
input, select, button {
    padding: 10px;
    margin: 8px;
}
button {
    background: #1d4ed8;
    color: white;
    border: 0;
}
#result {
    background: white;
    padding: 15px;
}
```

script.js

```
function calculate() {
    var amount = Number(document.getElementById("amount").value);
    var method = document.getElementById("method").value;
    if (amount <= 0) {
        document.getElementById("result").innerText = "Enter a valid amount.";
        return;
    }
    var fee = amount * 2 / 100;
    document.getElementById("result").innerText =
        method + " payment\nFee (2%): Rs. " + fee.toFixed(2) +
        "\nTotal: Rs. " + (amount + fee).toFixed(2) +
        "\nDemo only. Do not enter real card details.";
}
```

OUTPUT



RESULT

Thus, traditional and token-based payment methods were compared using a simple payment simulation.

EXPERIMENT 6

Map Customer Touchpoints in E-CRM using a Case Study

Date: _____

AIM

To identify customer touchpoints in an online fashion store and record customer interactions.

PROCEDURE

49. Display the five customer journey stages: Awareness, Consideration, Purchase, Support and Retention.
50. Create fields for customer name, interaction stage and notes.
51. Add Add Interaction and Clear List buttons.
52. Use JavaScript to add each interaction to the list and clear the list when required.

CODING

crm.html

```
<!DOCTYPE html>
<html>
<head>
<title>E-CRM Touchpoints</title>
<link rel="stylesheet" href="style.css">
</head>
<body>
<h1>E-CRM Customer Touchpoints</h1>
<p>Awareness - Customer sees an advertisement.</p>
<p>Consideration - Customer compares products.</p>
<p>Purchase - Customer places an order.</p>
<p>Support - Customer asks for help.</p>
<p>Retention - Store sends follow-up offers.</p>
<input id="customer" placeholder="Customer name">
<select id="stage"><option>Awareness</option><option>Consideration</option>
<option>Purchase</option><option>Support</option><option>Retention</option></select>
<input id="notes" placeholder="Interaction notes">
<button onclick="addTouchpoint()">Add Interaction</button>
<button onclick="clearEntries()">Clear List</button>
<ol id="log"></ol>
<script src="script.js"></script>
</body>
</html>
```

style.css

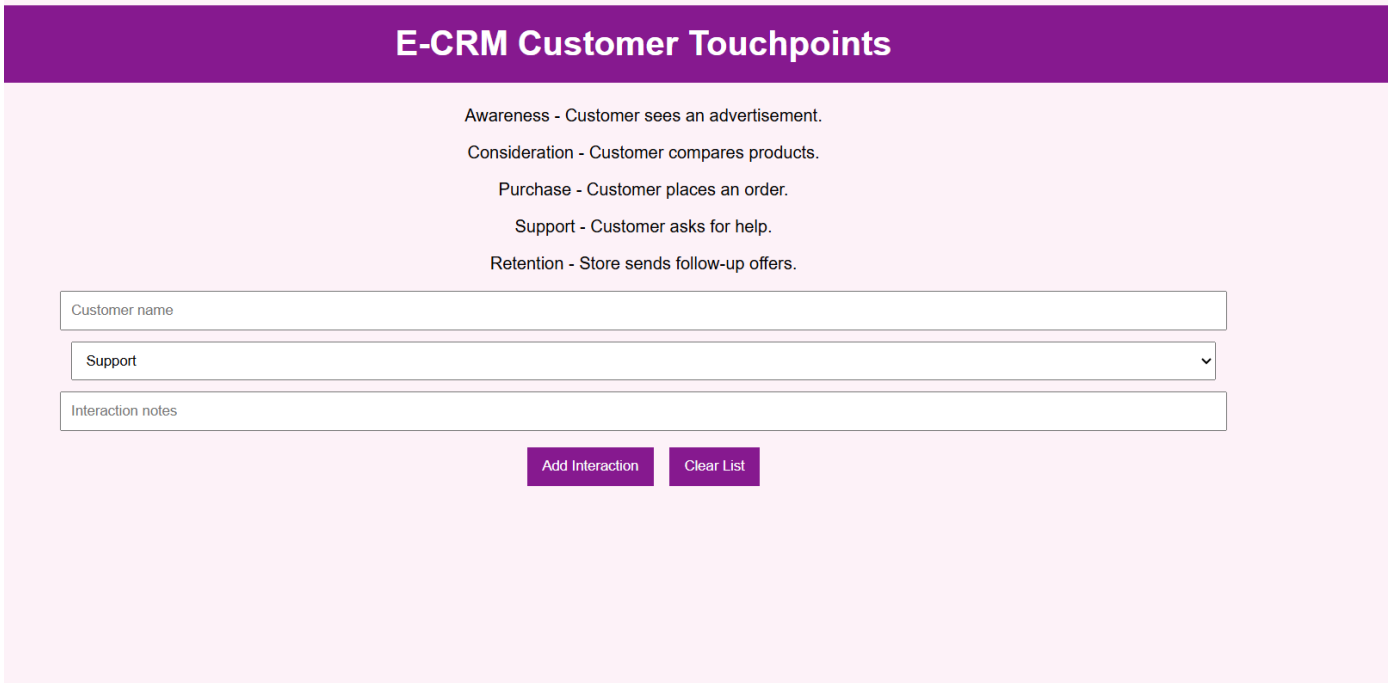
```
body {
    font-family: Arial;
    background: #fdf2f8;
    text-align: center;
}
h1 {
    background: #86198f;
    color: white;
    padding: 18px;
}
input, select {
    display: block;
    padding: 9px;
    margin: 10px auto;
```

```
    width: 70%;
}
button {
    padding: 10px 14px;
    margin: 5px;
    background: #86198f;
    color: white;
    border: 0;
}
li {
    background: #fae8ff;
    padding: 9px;
    margin: 8px auto;
    max-width: 650px;
    text-align: left;
}
```

script.js

```
function addTouchpoint() {
    var name = document.getElementById("customer").value;
    var stage = document.getElementById("stage").value;
    var notes = document.getElementById("notes").value;
    if (name == "" || notes == "") {
        alert("Enter customer name and notes."); return;
    }
    var item = document.createElement("li");
    item.innerText = name + " - " + stage + ": " + notes;
    document.getElementById("log").appendChild(item);
    document.getElementById("customer").value = "";
    document.getElementById("notes").value = "";
}
function clearEntries() {
    document.getElementById("log").innerHTML = "";
}
```

OUTPUT



RESULT

Thus, customer touchpoints were mapped in an online fashion store case study and interactions were recorded successfully.